Analysis / Polylogarithms / Transseries

Double Resonance and Nested Stokes Geometry in Polylogarithmic Lambert Inversion

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Abstract

We study the symmetric specialization $\mathcal{F}_\varepsilon(t) = \sum_{a,b\geq 1}(a/b)^\varepsilon e^{-abt}/(ab)$ of the established two-index Lambert family. Three Mellin poles coalesce at $\varepsilon = 0$, producing a quadratic logarithmic core. A centered Gamma–zeta normalization gives every resonant order jet and removes its next-to-leading logarithmic degree. The trivial zeros suppress every odd perturbative power beyond the linear term; the second order jet restores that entire parity sector, with an explicit arithmetic Borel kernel and a Stokes jump exactly $\pi^2/2$ times the unperturbed jump.

The principal inverse contribution is a uniform arcsinh crossover when $\varepsilon = \lambda/X$ and the target is $X^2/2$. We compute its first three corrections and the displacement of the two complex folds of its singular core. At exact resonance we give a finite formula for every inverse-sector coefficient, prove alternating coefficientwise signs, determine exact valuations and Catalan highest-degree terms, and prove zero ordinary radius but exact Borel radius $4\pi^2$ for every positive fixed slow parameter. An elementary Mellin proof fixes the exact real completion. Lagrange inversion then transports the entire divisor-weighted Stokes jump to the inverse. In the physical variable the jump is doubly exponentially small. The arithmetic average of the lateral inverses differs from the real inverse by a second-action term whose leading coefficient simplifies to -1. Every asserted theorem has a conventional proof; finite symbolic checks, numerical diagnostics, a scoped source audit, and further research questions accompany the article.

Status. Unrefereed mathematical research with explicit proofs; not a Lean formalization. The Lambert family and its general resurgent infrastructure are prior work. The explicit resonant inverse and jet refinements are proposed contributions, not certified claims of worldwide priority. No repository files were modified.

Contents

1 Research setting, antecedents, and principal results **3**
1.1 Why this particular intersection . . . 3
1.2 Prior work and the claimed scope of this contribution . . . 3
1.3 A preview of the exact identities . . . 3

2 The symmetric family and its Mellin representation **4**

3 Three coalescing poles and all centered order jets **5**
3.1 A centered finite formula for every logarithmic jet . . . 6

4 Arithmetic Borel kernels and second-jet parity restoration **7**
4.1 Reflection before differentiation . . . 7
4.2 An entire parity sector restored by the second jet . . . 8
4.3 Finite all-order harmonic and zeta-derivative coordinates . . . 9

5 An exact real completion and its arithmetic jump **9**

6 Uniform inverse crossover and the displaced core folds **10**
6.1 A convergent chart for the core, not for the full tail . . . 10
6.2 Transport from the core to the actual positive inverse . . . 11
6.3 Explicit critical-locus displacement for the complex core . . . 12

7 Exact inverse-sector coefficients: signs, Catalan degrees, and divergence **13**
7.1 Exact Borel radius of the inverse-sector series . . . 14
7.2 The actual inverse and finite-sector error control . . . 15

8 Exact inverse Stokes transport and the nested exponential scale **16**
8.1 A convergent Lagrange formula in the Stokes parameter . . . 16
8.2 The first two inverse Stokes effects . . . 17

9 Verification and reproducibility **18**
9.1 Exact checks . . . 18
9.2 Numerical diagnostics . . . 18
9.3 How to reproduce . . . 19

10 Source audit, corrections, and integration boundaries **19**
10.1 What was inspected . . . 19
10.2 No unsupported source-specific correction . . . 19
10.3 Recommended integration shape . . . 20

11 Further research questions **20**
11.1 Sharp large-order constants of the inverse sectors . . . 20
11.2 Full inverse Borel continuation . . . 20
11.3 Uniform Stokes transport away from exact resonance . . . 20
11.4 From core folds to full complex Lambert folds . . . 20
11.5 Character pairs and higher collision multiplicity . . . 21
11.6 Certified computation and formal proof organization . . . 21

Conclusion **21**

1 Research setting, antecedents, and principal results

1.1 Why this particular intersection

A single polylogarithm near its unit argument has a locally convergent power–log expansion after the appropriate pole cancellation. Summing polylogarithms on a multiplicative lattice changes the analytic class. The resulting Lambert-type function has a factorially divergent perturbative tail and an arithmetic lattice of Borel singularities. Nonlinear inversion then produces two distinct exponential levels: ordinary sectors of size e^{-nX} and Stokes sectors of size $\exp(-4\pi^2 e^{X-\gamma})$.

The distinction is consequential for the two ProveIt projects. The Polylogarithms sources develop normalized Gamma/zeta order bridges; the Transseries sources distinguish exact cores, formal reversion, analytic charts, and residual transport. This article connects those operations without importing a convergence assertion from the wrong analytic class. The inspected bridge chapter explicitly regularizes trivial zeros before taking logarithms, and the transseries overview explicitly distinguishes coefficient-form from operator-form Lagrange inversion [8, 9].

1.2 Prior work and the claimed scope of this contribution

The family considered here is not new. With trivial characters and $q = e^{-t}$ it is exactly

$$\mathcal{F}_\varepsilon(t) = \Xi_{1-\varepsilon,1+\varepsilon}(1,1;q)$$

in the notation of Broadhurst and Dorigoni [1], who treat arbitrary pairs of indices and Dirichlet characters and their full resurgent expansions. Their 2026 overview states this general family and its transformed nonperturbative terms explicitly [2]. Dorigoni and Kleinschmidt had already developed resurgent expansions of Lambert series and iterated Eisenstein integrals [3]. Accordingly, neither the family, the Mellin method, the existence of arithmetic Borel singularities, nor the general phenomenon of hidden exponential contributions is claimed here as a discovery.

Our narrower proposed refinements are a centered triple-collision jet calculus, a closed parity-jet Stokes identity, an arcsinh inverse crossover with explicit Gamma–Stieltjes corrections, a finite rational inverse coefficient formula with sharp structural consequences, and explicit nonlinear transport to a nested exponential scale. The formulas are proved directly, making their normalization and domains independently checkable. The exact scalar median identity is treated as a useful specialization of the established Lambert/Voronoi structure, not as a new general transformation formula.

The library counterparts of three recent research drafts were also compared at the level of their abstracts and selected relevant statements: the Gamma-normalized resonant Lerch jets, the uniform single-polylogarithm fold analysis, and the Fermi–Dirac/dilogarithm reversion package [11, 12, 13]. Their analytic tails and singular cores differ from the double Lambert sum here. We do not claim to have extracted or audited every member of the five incoming ZIP archives. Section 10 specifies the boundary of the inspection.

1.3 A preview of the exact identities

Write $d(n)$ for the number of positive divisors of n, and put

$$A = 4\pi^2, \qquad C_* = \frac{\zeta(2)}{2} - \gamma^2 - 2\gamma_1. \tag{1.1}$$

The real resonant function is

$$\mathcal{F}_0(t) = \sum_{n\geq 1} \frac{d(n)}{n} e^{-nt} = -\sum_{a\geq 1} \frac{\log(1-e^{-at})}{a}, \qquad t > 0.$$

Its exact Borel completion is

$$\mathcal{F}_0(t) = \frac{1}{2}(\log(1/t)+\gamma)^2 + C_* - \frac{t}{4} - 2\sum_{n\geq 1} d(n)\int_0^\infty e^{-\xi/t}\log\left|1-\frac{\xi^2}{A^2n^2}\right|\frac{\mathrm{d}\xi}{\xi}. \tag{1.2}$$

The two lateral values of the logarithm give a jump

$$J(t) = 4\pi\mathrm{i}\sum_{n\geq 1} d(n)E_1(An/t), \qquad E_1(x) = \int_x^\infty \frac{e^{-v}}{v}\,\mathrm{d}v. \tag{1.3}$$

If g_0 is the real inverse of $\mathcal{F}_0$, and $g_\pm$ are the nearby inverses of $\mathcal{F}_0 \pm J/2$, then, with $t = g_0(Y) \to 0^+$,

$$g_+(Y) - g_-(Y) \sim \frac{\mathrm{i}}{\pi}\,\frac{t^2}{\log(1/t)+\gamma}e^{-A/t}, \tag{1.4}$$

$$\frac{g_+(Y)+g_-(Y)}{2} - g_0(Y) \sim -\frac{t^2}{(\log(1/t)+\gamma)^2}e^{-2A/t}. \tag{1.5}$$

The coefficient -1 in the second line is an exact simplification, not a fitted numerical constant. Arithmetic averaging in that line is not being identified with the general nonlinear notion of median resummation.

2 The symmetric family and its Mellin representation

Definition 2.1. For $\Re t > 0$ and $\varepsilon \in \mathbb{C}$, define

$$D_\varepsilon(n) = \sum_{ab=n}(a/b)^\varepsilon, \tag{2.1}$$

$$\mathcal{F}_\varepsilon(t) = \sum_{a,b\geq 1}\frac{(a/b)^\varepsilon}{ab}e^{-abt} = \sum_{n\geq 1}\frac{D_\varepsilon(n)}{n}e^{-nt}. \tag{2.2}$$

The powers use real logarithms of positive integers. All logarithms of t below use the branch real on the positive axis, unless a different local continuation is explicitly specified.

Proposition 2.2 (Analyticity, symmetry, and the polylogarithm bridge)**.** *The family is jointly holomorphic on $\{\Re t > 0\}\times\mathbb{C}$, even in ε, and satisfies*

$$\mathcal{F}_\varepsilon(t) = \sum_{a\geq 1} a^{\varepsilon-1}\,\mathrm{Li}_{1+\varepsilon}(e^{-at}). \tag{2.3}$$

For real ε and $t > 0$ it is positive and strictly decreasing in t, with range $(0,\infty)$. For $c > |\Re\varepsilon|$ it has the Mellin formula

$$\mathcal{F}_\varepsilon(t) = \frac{1}{2\pi\mathrm{i}}\int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty}\Gamma(s)\zeta(1+s-\varepsilon)\zeta(1+s+\varepsilon)t^{-s}\,\mathrm{d}s. \tag{2.4}$$

Proof. On a compact parameter set with $|\Re\varepsilon| \leq M$, $|D_\varepsilon(n)| \leq d(n)n^M$. Exponential decay in n gives normal convergence, also after any fixed number of parameter or t derivatives. Interchanging a and b proves evenness. Expanding the defining polylogarithm series proves (2.3); that series normalization is classical [4]. For real ε, the summands are positive and their t derivatives are negative. The limit at infinity is zero. Pairing (a,b) with (b,a) gives $D_\varepsilon(n) \geq d(n)$, while the partial sum of $1/(ab)$ over $a,b \leq M$ tends to infinity as $M\to\infty$. Thus the limit at zero is infinite. Finally insert Mellin inversion for e^{-abt}. Absolute convergence on the line follows from $c > |\Re\varepsilon|$ and the two absolutely convergent Dirichlet series. The exponential decay of $\Gamma(c+\mathrm{i}v)$ completes the interchange and also permits $|\arg t| < \pi/2$. □

The Dirichlet series of the finite arithmetic weights is particularly simple:

$$\sum_{n\geq 1}\frac{D_\varepsilon(n)}{n^z}=\zeta(z-\varepsilon)\zeta(z+\varepsilon),\qquad \Re z>1+|\Re\varepsilon|. \tag{2.5}$$

For later use set

$$M_r(n)=\sum_{ab=n}\log^r(a/b). \tag{2.6}$$

Then $M_{2r+1}(n)=0$ and

$$\sum_{n\geq 1}\frac{M_{2r}(n)}{n^z}=\sum_{j=0}^{2r}(-1)^j\binom{2r}{j}\zeta^{(j)}(z)\zeta^{(2r-j)}(z),\qquad \Re z>1. \tag{2.7}$$

This exact identity expresses the order jets both in finite divisor moments and in ordinary zeta derivatives.

3 Three coalescing poles and all centered order jets

Use the Stieltjes convention

$$\zeta(1+z)=\frac{1}{z}+\sum_{r\geq 0}\frac{(-1)^r\gamma_r}{r!}z^r,\qquad \gamma_0=\gamma. \tag{3.1}$$

For $0<|\varepsilon|<1$ define the singular core

$$\mathcal{C}_\varepsilon(L)=\Gamma(\varepsilon)\zeta(1+2\varepsilon)e^{\varepsilon L}+\Gamma(-\varepsilon)\zeta(1-2\varepsilon)e^{-\varepsilon L}+\zeta(1-\varepsilon)\zeta(1+\varepsilon). \tag{3.2}$$

Each of the first two terms has a double pole in ε; the sum does not.

Theorem 3.1 (Uniform pole cancellation and complete power coefficients)**.** *The apparent singularity of $\mathcal{C}_\varepsilon(L)$ at $\varepsilon=0$ is removable, and the result is entire in L and holomorphic and even in $|\varepsilon|<1$. Its resonant value is*

$$\mathcal{C}_0(L)=\frac{1}{2}(L+\gamma)^2+C_*. \tag{3.3}$$

For every $N\geq 0$, $0<\rho<1$, and $\eta>0$,

$$\mathcal{F}_\varepsilon(t)=\mathcal{C}_\varepsilon(\log(1/t))+\sum_{k=1}^{N}r_k(\varepsilon)t^k+O(t^{N+1}),\qquad r_k(\varepsilon)=\frac{(-1)^k}{k!}\zeta(1+\varepsilon-k)\zeta(1-\varepsilon-k), \tag{3.4}$$

uniformly for $|\varepsilon|\leq\rho$ and $|\arg t|\leq\pi/2-\eta$ as $t\to 0$. On smaller parameter disks the same assertion can be differentiated any fixed number of times in ε.

Proof. The Mellin integrand has the three moving poles $s=\varepsilon,-\varepsilon,0$. Their residues give (3.2). A fixed circle $|s|=r$ with $\rho<r<1$ represents their sum by a contour integral. Its integrand is holomorphic in ε on a slightly larger disk and entire in L. This proves removability, uniformity, and symmetry without subtracting large singular quantities numerically.

At $\varepsilon=0$, multiply

$$\Gamma(s)=s^{-1}-\gamma+\tfrac{1}{2}(\gamma^2+\zeta(2))s+O(s^2),\qquad \zeta(1+s)^2=s^{-2}+2\gamma s^{-1}+\gamma^2-2\gamma_1+O(s).$$

The coefficients of s^{-3},s^{-2},s^{-1} in the product are $1,\gamma,\zeta(2)/2-\gamma^2/2-2\gamma_1$. Taking the residue after multiplication by e^{sL} gives (3.3).

Shift the Mellin line to $\Re s = -N-1-\delta$, with $0<\delta<1$. The additional poles crossed are the Gamma poles $s=-1,\dots,-N-1$, whose residues are exactly $r_k(\varepsilon)t^k$. Polynomial bounds for zeta in fixed vertical strips, together with Stirling's exponential decay and the prescribed t sector, give a uniform $O(t^{N+1+\delta})$ remaining integral. Absorbing the $k=N+1$ term proves the displayed remainder. Choose a larger compact parameter disk still inside $|\varepsilon|<1$ and apply Cauchy's formula to obtain the derivative assertions. The Gamma expansion and zeta functional identities used here have their standard normalizations [6, 5]. □

3.1 A centered finite formula for every logarithmic jet

Define the removable analytic functions

$$\mathfrak{a}(z)=z^2\Gamma(z)\zeta(1+2z)e^{-\gamma z}=\sum_{j\ge 0}a_jz^j,\qquad \mathfrak{z}(z)=z^2\zeta(1-z)\zeta(1+z)=\sum_{j\ge 0}z_jz^j. \tag{3.5}$$

Their initial coefficients are

$$a_0=\tfrac12,\qquad a_1=0,\qquad a_2=b:=\frac{\zeta(2)}{4}-\gamma^2-2\gamma_1, \tag{3.6}$$

$$a_3=\frac43\gamma^3+4\gamma\gamma_1+2\gamma_2-\frac{\zeta(3)}{6},\qquad z_0=-1,\qquad z_2=\delta_0:=\gamma^2+2\gamma_1. \tag{3.7}$$

All odd z_j vanish. The notation z_j is a coefficient label, not a zeta value.

Theorem 3.2 (Centered triple-resonance jets)**.** *Let $W=L+\gamma$. For every $r\ge 0$,*

$$\partial_\varepsilon^{2r}\mathcal{C}_\varepsilon(L)\Big|_{\varepsilon=0}=(2r)!\left\{2\sum_{j=0}^{2r+2}\frac{a_jW^{2r+2-j}}{(2r+2-j)!}+z_{2r+2}\right\}. \tag{3.8}$$

Its degree is exactly $2r+2$, its leading coefficient is $1/((2r+1)(2r+2))$, and the coefficient of W^{2r+1} is zero. Every odd order jet vanishes. Formula (3.5), together with the convergent log-Gamma series, is a finite coefficient algorithm in Stieltjes constants and zeta values for each fixed jet.

Proof. The centered identity is

$$\mathcal{C}_\varepsilon(L)=\frac{\mathfrak{a}(\varepsilon)e^{\varepsilon W}+\mathfrak{a}(-\varepsilon)e^{-\varepsilon W}+\mathfrak{z}(\varepsilon)}{\varepsilon^2}. \tag{3.9}$$

The numerator is even and its constant term is $2a_0+z_0=0$. Extracting ε^{2r+2} proves (3.8); $a_0=1/2$ and $a_1=0$ give the two highest-degree assertions. For completeness,

$$\mathfrak{a}(z)=\exp\left(-2\gamma z+\sum_{m\ge 2}\frac{(-1)^m\zeta(m)}{m}z^m\right)\left(\frac12+\gamma z-2\gamma_1z^2+2\gamma_2z^3+\cdots\right).$$

Multiplication gives (3.6)–(3.7) and computes any prescribed finite order using only finite Taylor data. □

Thus the centered second jet has the form

$$\partial_\varepsilon^2\mathcal{C}_\varepsilon(L)\Big|_0=\frac{W^4}{12}+2bW^2+4a_3W+4a_4+2z_4.$$

The missing cubic term is a structural consequence of the centering, not a numerical cancellation. Combining Theorems 3.1 and 3.2 gives a complete asymptotic expansion of $\sum_n M_{2r}(n)e^{-nt}/n$ at every fixed order $2r$.

4 Arithmetic Borel kernels and second-jet parity restoration

4.1 Reflection before differentiation

For $k \geq 2$ the zeta functional equation gives

$$r_k(\varepsilon) = \frac{2\{1 + (-1)^k \cos(\pi\varepsilon)\}}{A^k} \frac{\Gamma(k-\varepsilon)\Gamma(k+\varepsilon)}{k!} \zeta(k-\varepsilon)\zeta(k+\varepsilon). \tag{4.1}$$

Indeed, apply $\zeta(1-z) = 2(2\pi)^{-z}\cos(\pi z/2)\Gamma(z)\zeta(z)$ to $z = k-\varepsilon, k+\varepsilon$, then use the cosine product formula. The restriction $k \geq 2$, $|\varepsilon| < 1$ avoids the Dirichlet-series boundary at 1. The linear coefficient is retained separately:

$$r_1(\varepsilon) = -\zeta(\varepsilon)\zeta(-\varepsilon). \tag{4.2}$$

Definition 4.1 (Shifted Borel normalization)**.** For the formal tail $\mathcal{R}_\varepsilon(t) = \sum_{k\geq 2} r_k(\varepsilon)t^k$, put

$$\widehat{\mathcal{R}}_\varepsilon(\xi) = \sum_{k\geq 2} \frac{r_k(\varepsilon)}{\Gamma(k)} \xi^{k-1}. \tag{4.3}$$

Thus Laplace integration has no extra factor $1/t$: $\int_0^\infty e^{-\xi/t}\xi^{k-1}\,\mathrm{d}\xi = \Gamma(k)t^k$.

Theorem 4.2 (Normally convergent hypergeometric divisor kernel)**.** *Let*

$$H_\varepsilon(z) = \Gamma(1-\varepsilon)\Gamma(1+\varepsilon)\left\{z\,{}_2F_1(1-\varepsilon, 1+\varepsilon; 2; z) - z\right\}. \tag{4.4}$$

For $|\varepsilon| < 1$, the Borel germ has the representation

$$\widehat{\mathcal{R}}_\varepsilon(\xi) = \frac{2}{\xi} \sum_{n\geq 1} D_\varepsilon(n) \left\{ H_\varepsilon\left(\frac{\xi}{An}\right) + \cos(\pi\varepsilon) H_\varepsilon\left(-\frac{\xi}{An}\right)\right\}. \tag{4.5}$$

It converges normally, jointly in ε and ξ, on compact subsets of $|\varepsilon| < 1$ and $\mathbb{C} \setminus ((-\infty, -A] \cup [A, \infty))$, with the value at $\xi = 0$ filled in. Parameter differentiation is legitimate there. At resonance it reduces to the elementary logarithmic kernel

$$\widehat{\mathcal{R}}_0(\xi) = -\frac{2}{\xi} \sum_{n\geq 1} d(n) \log\left(1 - \frac{\xi^2}{A^2n^2}\right). \tag{4.6}$$

The Taylor radius of this germ is exactly A.

Proof. The coefficient of z^k, $k \geq 2$, in H_ε is

$$\frac{\Gamma(k-\varepsilon)\Gamma(k+\varepsilon)}{k!\,\Gamma(k)}.$$

Insert this coefficient into (4.5), apply (2.5), and compare with (4.1). This proves the germ identity. For large n, $H_\varepsilon(\xi/(An)) = O_K(n^{-2})$ uniformly on any stated compact set; also $|D_\varepsilon(n)| \leq d(n)n^\rho$ for some $\rho < 1$. The majorant $\sum d(n)n^{-2+\rho}$ converges. Finitely many remaining hypergeometric terms have their usual analytic branches away from their cuts. This proves normal convergence and the derivative claim.

At $\varepsilon = 0$, $z\,{}_2F_1(1, 1; 2; z) = -\log(1-z)$, giving $H_0(z) = -\log(1-z) - z$ and hence (4.6). The $n = 1$ term has a nonzero logarithmic singularity at $\xi = A$; all $n \geq 2$ terms are analytic nearby. The same holds at $-A$. The germ is analytic in $|\xi| < A$, proving the exact radius. □

4.2 An entire parity sector restored by the second jet

At $\varepsilon = 0$, formula (4.1) gives

$$r_1(0) = -\frac{1}{4}, \qquad r_{2m}(0) = \frac{4\Gamma(2m)}{2m\,A^{2m}}\zeta(2m)^2 > 0, \qquad r_{2m+1}(0) = 0 \quad (m \geq 1). \tag{4.7}$$

In particular the nonzero even coefficients grow factorially. The vanishing odd coefficients have a double, rather than simple, zero as functions of the order parameter.

Theorem 4.3 (Exact parity activation and its Stokes multiplier)**.** *For every odd $k \geq 3$,*

$$r_k''(0) = \frac{2\zeta'(1-k)^2}{k!} = \frac{2\pi^2}{A^k}\frac{\Gamma(k)^2}{k!}\zeta(k)^2 > 0. \tag{4.8}$$

The Borel transform of the restored odd-power part is

$$\widehat{\mathcal{J}}_{\mathrm{odd}}(\xi) := \sum_{\substack{k\geq 3\\ k\ \mathrm{odd}}} \frac{r_k''(0)}{\Gamma(k)}\xi^{k-1} = \frac{2\pi^2}{\xi}\sum_{n\geq 1} d(n)\left\{\operatorname{artanh}\left(\frac{\xi}{An}\right) - \frac{\xi}{An}\right\}. \tag{4.9}$$

For the lateral boundary values continued from zero,

$$\operatorname{Disc}\widehat{\mathcal{R}}_0(\xi) = \frac{4\pi\mathrm{i}}{\xi}\sum_{An<\xi} d(n), \qquad \xi > 0, \tag{4.10}$$

$$\operatorname{Disc}\widehat{\mathcal{J}}_{\mathrm{odd}}(\xi) = \frac{2\pi^3\mathrm{i}}{\xi}\sum_{An<\xi} d(n) = \frac{\pi^2}{2}\operatorname{Disc}\widehat{\mathcal{R}}_0(\xi). \tag{4.11}$$

Consequently the corresponding lateral Laplace jumps satisfy the exact identity

$$(\mathcal{S}_+ - \mathcal{S}_-)\mathcal{J}_{\mathrm{odd}}(t) = 2\pi^3\mathrm{i}\sum_{n\geq 1} d(n)E_1(An/t) = \frac{\pi^2}{2}(\mathcal{S}_+ - \mathcal{S}_-)\mathcal{R}_0(t), \qquad t > 0. \tag{4.12}$$

Proof. For odd $k \geq 3$, the factor in braces in (4.1) is $1 - \cos(\pi\varepsilon)$, whose second derivative at zero is π^2. The other factors are nonzero and analytic there. This proves the second expression in (4.8). Alternatively, $\zeta(1-k+\varepsilon)\zeta(1-k-\varepsilon) = -\zeta'(1-k)^2\varepsilon^2 + O(\varepsilon^4)$, and $(-1)^k = -1$; this proves the first expression, including its sign.

After division by $\Gamma(k)$ the coefficient is $2\pi^2\zeta(k)^2/(kA^k)$. The identity $\sum_{k\geq 3,\ k\,\mathrm{odd}} z^k/k = \operatorname{artanh} z - z$, followed by $\zeta(k)^2 = \sum d(n)n^{-k}$, proves (4.9). All interchanges first take place in $|\xi| < A$; the same large-n estimates as before extend them to compact slitted domains.

We fix the signs explicitly. For $\xi > An$, approaching from the upper half-plane makes $1-\xi^2/(A^2n^2)$ approach the negative axis from below. Thus its logarithm has upper-minus-lower jump $-2\pi\mathrm{i}$. The minus sign in (4.6) gives (4.10). Similarly $\operatorname{artanh} z = (\log(1+z) - \log(1-z))/2$ has jump $\pi\mathrm{i}$ for $z > 1$. This gives (4.11). Integrating each step function against $e^{-\xi/t}/\xi$ produces $E_1(An/t)$. The divisor sum converges exponentially, justifying the integration. □

Remark 4.4. "Odd-power part" refers to the original powers t^k, not to parity in the Borel coordinate: the shifted transform uses ξ^{k-1}. The theorem concerns this projected part of the second jet. It does not assert that the *whole* second order jet is $\pi^2/2$ times the original function.

4.3 Finite all-order harmonic and zeta-derivative coordinates

The regular Gamma product in (4.1) satisfies

$$\frac{\Gamma(k-\varepsilon)\Gamma(k+\varepsilon)}{\Gamma(k)^2} = \exp\left(\sum_{j\geq 1} \frac{\zeta(2j) - H_{k-1}^{(2j)}}{j}\varepsilon^{2j}\right), \qquad H_m^{(r)} = \sum_{\ell=1}^{m} \ell^{-r}. \tag{4.13}$$

This follows by adding the Taylor series of the two log-Gamma factors; $\psi^{(2j-1)}(k) = (2j-1)!(\zeta(2j) - H_{k-1}^{(2j)})$. Consequently, for every $r \geq 0$ and $k \geq 2$,

$$r_k^{(2r)}(0) = \frac{2\Gamma(k)^2(2r)!}{k!A^k}[\varepsilon^{2r}]\Bigg(\{1+(-1)^k\cos(\pi\varepsilon)\}\exp\left(\sum_{j=1}^{r}\frac{\zeta(2j)-H_{k-1}^{(2j)}}{j}\varepsilon^{2j}\right) \times \sum_{\ell=0}^{r}\frac{\varepsilon^{2\ell}}{(2\ell)!}\sum_{j=0}^{2\ell}(-1)^j\binom{2\ell}{j}\zeta^{(j)}(k)\zeta^{(2\ell-j)}(k)\Bigg). \tag{4.14}$$

Only finitely many coefficients are used at each prescribed order. Together with (3.8), this supplies both the logarithmic core and every power coefficient of every symmetric order derivative. No differentiation of a singular logarithm at a trivial zero is needed.

5 An exact real completion and its arithmetic jump

The existence of a real function with a prescribed asymptotic expansion does not, by itself, fix any beyond-all-orders real term. We therefore prove the equality to the specified Lambert function, rather than infer it from reality or conjugacy of lateral values.

Theorem 5.1 (Exact logarithmic Borel completion)**.** *For every $t > 0$, define*

$$\mathcal{S}_{\pm}\mathcal{R}_0(t) = \int_0^{\infty} e^{-\xi/t}\widehat{\mathcal{R}}_0(\xi \pm \mathrm{i}0)\,\mathrm{d}\xi.$$

The integrals exist, with integrable logarithmic boundary singularities, and

$$\mathcal{F}_0(t) = \mathcal{C}_0(\log(1/t)) - \frac{t}{4} + \frac{1}{2}\{\mathcal{S}_+\mathcal{R}_0(t) + \mathcal{S}_-\mathcal{R}_0(t)\}, \tag{5.1}$$

$$\mathcal{S}_+\mathcal{R}_0(t) - \mathcal{S}_-\mathcal{R}_0(t) = J(t) = 4\pi\mathrm{i}\sum_{n\geq 1} d(n)E_1(An/t). \tag{5.2}$$

Thus the forward lateral functions are exactly

$$\mathcal{F}_{\pm}(t) := \mathcal{C}_0(\log(1/t)) - t/4 + \mathcal{S}_{\pm}\mathcal{R}_0(t) = \mathcal{F}_0(t) \pm J(t)/2. \tag{5.3}$$

Proof. First note an integrability estimate. For fixed $t > 0$ and $a \geq 1$,

$$\int_0^{\infty} e^{-\xi/t}\left|\log\left|1-\frac{\xi^2}{a^2}\right|\right|\frac{\mathrm{d}\xi}{\xi} \leq \frac{K_t}{a^2}.$$

On $\xi \leq a/2$ use $|\log(1-v)| \leq Cv$ for $0 \leq v \leq 1/4$. On $\xi > a/2$ substitute $\xi = au$ and retain half the exponential for integrability; the remaining factor is exponentially small in a. The logarithmic singularity at $u = 1$ is integrable. The imaginary step part has an even smaller exponential integral. Summation over $a = An$ is therefore justified by $\sum d(n)n^{-2} < \infty$.

It remains to identify the real part. The elementary Mellin identity

$$\int_0^{\infty} u^{-w-1}\log|1-u^2|\,\mathrm{d}u = \frac{\pi}{w}\cot\frac{\pi w}{2}, \qquad 0 < \Re w < 2, \tag{5.4}$$

follows by integration by parts, interpreted symmetrically at $u = 1$, and the substitution $v = u^2$. In detail, the derivative integral is $-w^{-1}\,\mathrm{PV}\int_0^\infty v^{-w/2}/(1-v)\,\mathrm{d}v$; $\mathrm{PV}\int_0^\infty v^{a-1}/(1-v)\,\mathrm{d}v = \pi\cot(\pi a)$ gives (5.4). The latter identity also follows by splitting at 1 and using the digamma reflection formula.

For

$$K(x) = -2\int_0^\infty e^{-u/x}\log|1-u^2|\frac{\mathrm{d}u}{u},$$

Mellin inversion of the exponential and (5.4) imply

$$K(x) = \frac{1}{2\pi\mathrm{i}}\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty} -\frac{2\pi}{w}\cot\frac{\pi w}{2}\Gamma(w)x^w\,\mathrm{d}w, \qquad 0 < b < 2. \tag{5.5}$$

Absolute integration in the intermediate Mellin variable follows from $0 < b < 2$ and Gamma decay, so this step does not rely on a formal exchange. Choose $1 < b < 2$, sum $d(n)K(t/(An))$, and apply $\sum d(n)n^{-w} = \zeta(w)^2$. The real average in (5.1) is therefore

$$\frac{1}{2\pi\mathrm{i}}\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty} -\frac{2\pi}{w}\cot\frac{\pi w}{2}\Gamma(w)\zeta(w)^2(t/A)^w\,\mathrm{d}w. \tag{5.6}$$

On the other hand, shift (2.4) for $\varepsilon = 0$ to $\Re s = -b$. The poles crossed give $\mathcal{C}_0(\log(1/t)) - t/4$. The functional equation yields

$$\Gamma(s)\zeta(1+s)^2 = -\frac{2\pi}{s}\cot\frac{\pi s}{2}\Gamma(-s)\zeta(-s)^2A^s.$$

Putting $w = -s$, including the orientation change, turns the remaining integral precisely into (5.6). This proves the exact identity, including any possible flat real term. Equation (5.2) is Theorem 4.3 integrated once. □

Corollary 5.2 (Arithmetic Stokes data)**.** *The positive Borel singularities lie at An, $n \geq 1$. The logarithmic jump accumulates the divisor counting function, and its Laplace image is an absolutely convergent sum of exponential integrals. In particular,*

$$J(t) = \frac{4\pi\mathrm{i}}{A}\,te^{-A/t}(1+O(t)) \qquad (t \to 0^+). \tag{5.7}$$

Proof. The local singularities and jump were determined explicitly. Integration by parts in $E_1(A/t)$ gives its first term and an $O(t)$ relative remainder. Every $n \geq 2$ term is exponentially smaller; the bound $d(n) \leq n$ is already sufficient to sum that tail. □

6 Uniform inverse crossover and the displaced core folds

6.1 A convergent chart for the core, not for the full tail

We use a scaled target $Y = X^2/2$ and set

$$u = X^{-1}, \qquad \varepsilon = \lambda u, \qquad w = u(L+\gamma). \tag{6.1}$$

The scaled core equation is $G(w, \lambda, u) = 0$, where

$$G(w,\lambda,u) = \frac{\mathfrak{a}(\lambda u)e^{\lambda w} + \mathfrak{a}(-\lambda u)e^{-\lambda w} + \mathfrak{z}(\lambda u)}{\lambda^2} - \frac{1}{2}. \tag{6.2}$$

The value at $\lambda = 0$ is removable. The cancellation $a_1 = 0$ is what eliminates a term linear in u.

Theorem 6.1 (Arcsinh crossover and its first corrections)**.** *Define the branches*

$$h(\lambda) = \frac{2\,\mathrm{arsinh}(\lambda/2)}{\lambda}, \quad h(0) = 1, \qquad s(\lambda) = \sqrt{1 + \lambda^2/4}, \quad s(0) = 1. \tag{6.3}$$

On every compact subset of $|\lambda| < 2$, the core inverse has a holomorphic chart

$$L_c = \frac{h(\lambda)}{u} - \gamma + uB(u, \lambda), \tag{6.4}$$

for sufficiently small u, with a genuinely convergent Taylor series in u. Its first two coefficients are

$$B(0, \lambda) = -\frac{C_* + b\lambda^2}{s(\lambda)}, \qquad \partial_u B(0, \lambda) = -2a_3\lambda^2. \tag{6.5}$$

Equivalently,

$$L_c = Xh(\lambda) - \gamma - \frac{C_* + b\lambda^2}{X\sqrt{1 + \lambda^2/4}} - \frac{2a_3\lambda^2}{X^2} + O(X^{-3}). \tag{6.6}$$

For real λ, the same assertion holds uniformly on any fixed compact interval, using analytic neighborhoods of that interval.

Proof. At $u = 0$,

$$G(w, \lambda, 0) = \frac{\cosh(\lambda w) - 1}{\lambda^2} - \frac{1}{2}.$$

The branch $w = h(\lambda)$ solves this equation because $\cosh(2\,\mathrm{arsinh}(\lambda/2)) = 1 + \lambda^2/2$. Its w derivative is $\sinh(\lambda h)/\lambda = s(\lambda)$, which does not vanish in $|\lambda| < 2$ or at any real λ. The holomorphic implicit function theorem therefore gives a unique chart $w(u, \lambda)$ locally along this branch. Uniqueness glues the local charts on compact sets. The absence of a linear u term in (6.2) implies $w = h + u^2 B$.

Writing $c_\lambda = 1 + \lambda^2/2$, the expansion is

$$\begin{aligned} G(w, \lambda, u) = \frac{\cosh(\lambda w) - 1}{\lambda^2} - \frac{1}{2} + u^2\{2b\cosh(\lambda w) + \delta_0\} \\ + 2a_3\lambda u^3 \sinh(\lambda w) + O(u^4). \end{aligned}$$

Since $2b + \delta_0 = C_*$, the u^2 coefficient gives $sB(0, \lambda) + C_* + b\lambda^2 = 0$. The u^3 coefficient gives $s\partial_u B(0, \lambda) + 2a_3\lambda \sinh(\lambda h) = 0$. Using $\sinh(\lambda h) = \lambda s$ proves (6.5) and (6.6). For a real compact interval $s(\lambda) > 0$ throughout, so the same argument supplies uniform neighborhoods and estimates. □

Every higher coefficient can be computed recursively from the analytic equation (6.2). For example, if $w = h + B_0 u^2 + B_1 u^3 + B_2 u^4 + \cdots$, then

$$B_2 = -\frac{\frac{1}{2}c_\lambda B_0^2 + 2b\lambda^2 s(\lambda) B_0 + \lambda^2(2a_4 c_\lambda + z_4)}{s(\lambda)}. \tag{6.7}$$

At $\lambda = 0$ this reduces to $B_2 = -C_*^2/2$, as required by $L_c = \sqrt{X^2 - 2C_*} - \gamma$.

6.2 Transport from the core to the actual positive inverse

Theorem 6.2 (First exponential correction in the crossover window)**.** *Fix a compact real interval of λ values. Let $\varepsilon = \lambda/X$, let L_c be the core root of Theorem 6.1, and put $t_c = e^{-L_c}$. The unique positive solution of $\mathcal{F}_\varepsilon(t) = X^2/2$ has $L = \log(1/t)$ satisfying*

$$L - L_c = \frac{\zeta(\varepsilon)\zeta(-\varepsilon)}{\partial_L \mathcal{C}_\varepsilon(L_c)} t_c + O(t_c^2/X), \tag{6.8}$$

uniformly as $X \to \infty$. Hence

$$t = t_c - \frac{\zeta(\varepsilon)\zeta(-\varepsilon)}{\partial_L \mathcal{C}_\varepsilon(L_c)} t_c^2 + O(t_c^3/X). \tag{6.9}$$

The denominator is $Xs(\lambda) + O(X^{-1})$.

Proof. Differentiate the scaled core equation with respect to L: $\partial_L \mathcal{C}_\varepsilon(L_c) = XG_w(w_c, \lambda, u)$. The chart gives $G_w = s(\lambda) + O(u^2)$, uniformly bounded away from zero on the real compact set. Also $\partial_L^2 \mathcal{C}_\varepsilon$ is uniformly bounded there. By Theorem 3.1,

$$\mathcal{F}_\varepsilon(e^{-L}) = \mathcal{C}_\varepsilon(L) + r_1(\varepsilon)e^{-L} + O(e^{-2L}).$$

The derivative in L is positive and comparable to X near L_c; this follows either from the differentiated Mellin remainder or from the positive defining sum. A residual estimate first gives $L - L_c = O(t_c/X)$. Substitute this estimate back into the equation and Taylor expand the core and e^{-L} once. Then

$$0 = \partial_L \mathcal{C}_\varepsilon(L_c)(L - L_c) + r_1(\varepsilon)t_c + O(t_c^2).$$

Use (4.2) and divide by the derivative. Exponentiation gives (6.9); the square of $O(t_c/X)$ is absorbed by the stated remainder. Positivity and uniqueness are supplied by Proposition 2.2. □

A finite $1/X$ truncation of L_c does *not* isolate the exponential correction: its omitted algebraic error is much larger. Formula (6.8) uses the entire analytic core root, not a short truncation of (6.6).

6.3 Explicit critical-locus displacement for the complex core

Theorem 6.3 (Two moving simple folds of the normalized core)**.** *Set*

$$D = \delta_0 - 2b = 3\gamma^2 + 6\gamma_1 - \frac{\zeta(2)}{2}. \tag{6.10}$$

Near $(w, \lambda, u) = (\pi/2, \pm 2\mathrm{i}, 0)$ there are unique analytic solutions of $G = G_w = 0$. They satisfy

$$\lambda_\pm(u) = \pm 2\mathrm{i}\{1 + Du^2\} + O(u^4), \tag{6.11}$$

$$w_\pm(u) = \frac{\pi}{2}\{1 - Du^2\} + 8a_3u^3 + O(u^4). \tag{6.12}$$

Each remains a simple fold: locally, the two core inverse branches have a square-root interchange as λ encircles the critical value. This assertion concerns the singular core $\mathcal{C}_\varepsilon$, not the full Lambert function at these complex scalings.

Proof. At the positive imaginary point,

$$G = G_w = 0, \quad G_{ww} = -1, \quad G_\lambda = \mathrm{i}/2, \quad G_{w\lambda} = \mathrm{i}\pi/4.$$

The Jacobian of (G, G_w) with respect to (λ, w) is nonzero. The implicit function theorem gives the critical locus. The u^2 term of G at the point is $\delta_0 - 2b = D$, so $(\mathrm{i}/2)(\lambda_+(u) - 2\mathrm{i}) + Du^2 = O(u^3)$. This gives the u^2 coefficient of (6.11). The u^2 coefficient in $G_w = 0$ gives $w_+(u) - \pi/2 = -\pi Du^2/2 + O(u^3)$.

The order-u^3 term in G is proportional to $\sinh(\lambda w)$ and vanishes at the limiting fold; hence λ_+ has no u^3 term. Its w derivative at that point is $2a_3\lambda^2\cosh(\lambda w) = 8a_3$, giving the displayed $8a_3u^3$ term in w_+. Evenness of G in λ gives the negative branch. Finally $G_{ww} \neq 0$ and $G_\lambda \neq 0$ persist. Taylor expansion and the Weierstrass preparation theorem reduce the equation to a nondegenerate quadratic in the local w coordinate; its discriminant has a simple zero in λ. That is precisely the asserted square-root interchange. □

The fold location $\lambda = \pm 2\mathrm{i}$ is distinct from the secondary resonances $2\pi\mathrm{i}k$ in the single-polylogarithm crossover discussed in the earlier draft [12]. The distinction follows from the different leading functions: $\cosh(\lambda w) - 1$ here, rather than $(1 - e^{-\lambda})/\lambda$ there.

7 Exact inverse-sector coefficients: signs, Catalan degrees, and divergence

At exact resonance it is preferable to absorb the constant C_* into the large variable. In this section only, set

$$X = \sqrt{2(Y - C_*)}, \qquad u = X^{-1}, \qquad q = e^{\gamma - X}, \qquad L = X - \gamma + V. \tag{7.1}$$

This X differs from the scaled-target variable of Section 6; both definitions are explicitly tied to their targets. The positive branch is used, and Y is sufficiently large. Then $t = qe^{-V}$. Define the formal series including the linear term

$$R(q) = \sum_{n \geq 1} r_n q^n, \qquad r_n = r_n(0). \tag{7.2}$$

The formal inverse equation is

$$V + \frac{u}{2}V^2 + uR(qe^{-V}) = 0. \tag{7.3}$$

Here q and u are independent formal variables for coefficient extraction. They are specialized to the physical relation (7.1) only afterward.

Theorem 7.1 (Finite coefficient formula and sharp polynomial structure)**.** *There is a unique solution*

$$V(q, u) = \sum_{n \geq 1} C_n(u) q^n \in q\mathbb{Q}[u][[q]] \tag{7.4}$$

of (7.3)*. For every $n \geq 1$ it is given by*

$$C_n(u) = -\sum_{k=1}^{n} \frac{u^k}{k!}[q^n]R(q)^k \sum_{j=0}^{k-1} \binom{k-1}{j} n^{k-1-j} (k)_j \left(\frac{u}{2}\right)^j. \tag{7.5}$$

Its structural properties are

$$\deg C_n = 2n - 1, \qquad [u^{2n-1}]C_n = \frac{(-1)^{n+1}\,\mathrm{Cat}_{n-1}}{2^{3n-1}}, \tag{7.6}$$

$$(-1)^{n+1} C_n(u) \in \mathbb{Q}_{\geq 0}[u] \setminus \{0\}, \tag{7.7}$$

$$\mathrm{ord}_{u=0}\, C_n = \begin{cases} 1, & n = 1 \text{ or } n \text{ even}, \\ 2, & n \geq 3 \text{ odd}, \end{cases} \tag{7.8}$$

$$[u]C_n = -r_n, \qquad [u^2]C_n = \frac{n}{4} r_{n-1} \quad (n \geq 3 \text{ odd}). \tag{7.9}$$

In (7.7) *every nonzero coefficient is strictly positive after the indicated sign change. In particular every inverse sector is present, although every odd forward coefficient beyond the linear term vanishes.*

Proof. The constant term in q must be $V = 0$; the derivative with respect to V at $q = V = 0$ is 1. Equating coefficients recursively gives a unique solution over $\mathbb{Q}[u]$, since each r_n is rational.

For an explicit formula introduce an auxiliary variable η and write

$$V = -\eta u \frac{R(qe^{-V})}{1 + uV/2}.$$

Formal Lagrange inversion in η yields

$$V = \sum_{k\geq 1} \frac{(-\eta u)^k}{k!} \left(\frac{\partial}{\partial v}\right)^{k-1} \left\{R(qe^{-v})^k (1 + uv/2)^{-k}\right\}\Big|_{v=0}. \tag{7.10}$$

Because R has positive q valuation, only $k \leq n$ contributes to q^n. Thus setting $\eta = 1$ is legitimate coefficientwise, even though no analytic convergence of R is being assumed. For the q^n coefficient, $[q^n]R(qe^{-v})^k = e^{-nv}[q^n]R(q)^k$. Leibniz' rule gives

$$\partial_v^{k-1}\{e^{-nv}(1 + uv/2)^{-k}\}\Big|_0 = (-1)^{k-1} \sum_{j=0}^{k-1} \binom{k-1}{j} n^{k-1-j} (k)_j (u/2)^j.$$

Combining signs proves (7.5).

Since $k + j \leq 2k - 1 \leq 2n - 1$, the degree is at most $2n - 1$. The only term of that degree has $k = n$, $j = n - 1$, and uses $[q^n]R^n = r_1^n = (-1/4)^n$. Hence its coefficient is

$$-\frac{(-1/4)^n}{n!} \frac{(n)_{n-1}}{2^{n-1}} = \frac{(-1)^{n+1}}{2^{3n-1}} \frac{(2n-2)!}{n!(n-1)!},$$

which is exactly (7.6), and is nonzero.

For the sign assertion, every nonzero monomial contributing to $[q^n]R^k$ consists of factors $r_1 < 0$ and $r_{2m} > 0$. The number of r_1 factors has the same parity as n. Consequently $(-1)^n[q^n]R^k \geq 0$. Every coefficient of the inner polynomial in (7.5) is nonnegative, so no cancellation can reverse this sign. The degree computation ensures nonvanishing. The coefficient of u comes only from $k = 1$, and is $-r_n$. For odd $n \geq 3$ this vanishes. The coefficient of u^2 comes only from $k = 2, j = 0$ and equals $-n[q^n]R^2/2$. For odd $n \geq 3$, $[q^n]R^2 = 2r_1 r_{n-1} = -r_{n-1}/2$. This proves (7.9) and the exact valuations. □

The first three coefficients, useful as normalization checks, are

$$C_1(u) = \frac{u}{4}, \tag{7.11}$$

$$C_2(u) = -\frac{u}{288} - \frac{u^2}{16} - \frac{u^3}{32}, \tag{7.12}$$

$$C_3(u) = \frac{u^2}{384} + \frac{7u^3}{288} + \frac{3u^4}{128} + \frac{u^5}{128}. \tag{7.13}$$

The supplied exact verifier independently substitutes all coefficients through q^8 into (7.3). The proof, rather than that finite computation, establishes the all-order assertions.

7.1 Exact Borel radius of the inverse-sector series

Theorem 7.2 (Gevrey order and sharp Borel radius)**.** *For every $U > 0$, the polynomials in Theorem 7.1 satisfy, uniformly for $|u| \leq U$,*

$$|C_n(u)| \leq K_U A^{-n} \Gamma(n) e^{K_U \sqrt{n}}. \tag{7.14}$$

For every fixed real $u > 0$, the ordinary series $\sum C_n(u) q^n$ has radius zero, whereas its shifted Borel transform

$$\widehat{V}_u(\xi) = \sum_{n\geq 1} \frac{C_n(u)}{\Gamma(n)} \xi^{n-1} \tag{7.15}$$

has radius exactly $A = 4\pi^2$. In particular it is a Gevrey-one series, but not a convergent exponential-sector chart at fixed $u > 0$. The point $\xi = -A$ is a singularity of the Borel germ.

Proof. Formula (4.7) gives a constant $K > 0$ with $|r_m| \le KA^{-m}(m-1)!$ for all $m \ge 1$. For a composition $m_1 + \cdots + m_k = n$ with $m_i \ge 1$, $\prod_i (m_i - 1)! \le (n-k)!$. Counting the compositions gives

$$|[q^n]R(q)^k| \le K^k A^{-n}(n-k)!\binom{n-1}{k-1} = K^k A^{-n}\frac{(n-1)!}{(k-1)!}. \tag{7.16}$$

Also $(k)_j \le (2n)^j$ for $0 \le j \le k-1$, so the absolute value of the inner sum in (7.5) is at most $n^{k-1}(1+U)^{k-1}$. Therefore

$$|C_n(u)| \le A^{-n}\Gamma(n)KU\sum_{j=0}^{n-1}\frac{\{KUn(1+U)\}^j}{(j+1)!j!}.$$

The elementary bound $\sum_{j\ge 0} x^j/(j!)^2 \le e^{2\sqrt{x}}$ for $x \ge 0$ follows by comparison with the product $e^{\sqrt{x}}e^{\sqrt{x}}$. This proves (7.14). For any $A' < A$, its right-hand side is at most $K_{U,A'}(A')^{-n}\Gamma(n)$, which is the usual Gevrey-one estimate. In particular the Borel radius is at least A.

Fix $u > 0$. For even n, all terms in $C_n(u)$ have the same negative sign, and the $k = 1$ term gives

$$|C_n(u)| \ge ur_n = \frac{4u\Gamma(n)}{nA^n}\zeta(n)^2.$$

This lower bound proves radius zero for the ordinary series and Borel radius at most A. Thus the latter radius is exactly A. Finally (7.7) shows that $\widehat{V}_u(-\xi)$ has nonnegative Taylor coefficients. A power series with nonnegative coefficients and finite radius A cannot be analytic at its positive endpoint A: otherwise its Taylor expansion about a point just below A, with nonnegative derivatives, can be rearranged to make the original series converge at a point beyond A. This contradiction is the elementary positive-coefficient form of Pringsheim's argument. Applied to $\widehat{V}_u(-\xi)$, it proves singularity at $-A$. □

Remark 7.3 (What the sharp radius does not prove). The theorem locates the nearest guaranteed negative-axis singularity and fixes the radius of the Borel germ. It does not, by itself, prove global analytic continuation of the inverse Borel transform, classify all its singularities, or identify every alien derivative. The lateral inverse functions in Section 8 are defined analytically from the *exact forward lateral functions*, so their jump does not depend on those stronger unproved assertions.

7.2 The actual inverse and finite-sector error control

Theorem 7.4 (Physical specialization of the formal coefficients)**.** *Let $t = g_0(Y)$, with the physical variables (7.1), and let $V = \log(q/t)$. For every fixed $N \ge 1$,*

$$V = \sum_{n=1}^{N} C_n(1/X)q^n + O(q^{N+1}/X), \qquad X \to +\infty. \tag{7.17}$$

In particular,

$$t = q\left(1 - \frac{q}{4X} + O(q^2/X)\right), \qquad \frac{A}{t} = \frac{A}{q} + \frac{A}{4X} + O(q/X). \tag{7.18}$$

Proof. The exact expansion (3.4) at $\varepsilon = 0$, with a fixed finite remainder order, gives

$$0 = XV + \frac{1}{2}V^2 + \sum_{m=1}^{N} r_m q^m e^{-mV} + O(q^{N+1})$$

near $V = 0$. Monotonicity first gives $V = O(q/X)$, for example by comparing the residual at $V = 0$ with the derivative, which is $X + O(q)$. Substitute $V_N = \sum_{n \le N} C_n(1/X)q^n$. The formal coefficient identity cancels every power through q^N. Since N is fixed and the coefficients are polynomials bounded as $1/X \to 0$, the remaining residual is $O(q^{N+1})$. The derivative of the exact equation is bounded below by $X/2$ in this neighborhood, giving the stated $O(q^{N+1}/X)$ correction. Using $C_1 = 1/(4X)$ and exponentiating proves (7.18). □

This is an asymptotic statement at every fixed sector order. It is not an assertion that the infinite divergent series may be evaluated at the physical pair $(q, 1/X)$ by ordinary summation.

8 Exact inverse Stokes transport and the nested exponential scale

The forward jump in (5.2) is explicit. We now transport it through functional inversion without silently linearizing away the higher nonlinear sectors.

8.1 A convergent Lagrange formula in the Stokes parameter

Let g_0 denote the positive inverse of $\mathcal{F}_0$. For Y sufficiently large, let $g_\eta(Y)$ be the local inverse near $t = g_0(Y)$ of

$$\mathcal{F}_\eta(z) = \mathcal{F}_0(z) + \frac{\eta}{2} J(z). \tag{8.1}$$

The analytic continuations used here lie in a small disk around positive t; $E_1(An/z)$ uses the continuation from positive z. For $\eta = \pm 1$ this agrees with the two forward lateral functions.

Theorem 8.1 (All-order analytic transport of the arithmetic jump)**.** *For sufficiently large real Y, the inverse $g_\eta(Y)$ is holomorphic in η on a disk of radius greater than 1, and*

$$g_\eta(Y) = g_0(Y) + \sum_{k \ge 1} \frac{(-\eta)^k}{k!} \partial_Y^{k-1} \left[g_0'(Y) \left\{ \frac{J(g_0(Y))}{2} \right\}^k \right]. \tag{8.2}$$

The series is actually convergent at $\eta = \pm 1$. Consequently

$$g_+(Y) - g_-(Y) = -2 \sum_{j \ge 0} \frac{1}{(2j+1)!} \partial_Y^{2j} \left[g_0'(Y) \left\{ \frac{J(g_0(Y))}{2} \right\}^{2j+1} \right], \tag{8.3}$$

$$\frac{g_+(Y) + g_-(Y)}{2} - g_0(Y) = \sum_{j \ge 1} \frac{1}{(2j)!} \partial_Y^{2j-1} \left[g_0'(Y) \left\{ \frac{J(g_0(Y))}{2} \right\}^{2j} \right]. \tag{8.4}$$

These formulas retain the entire arithmetic divisor sum in J, not only its first action.

Proof. First establish a domain in which the series is more than formal. Let $t = g_0(Y)$ and $W = \log(1/t) + \gamma$. On a disk $|z - t| \le ct^2$, with fixed sufficiently small $c > 0$, the differentiated Mellin expansion gives

$$\mathcal{F}_0'(z) = -\frac{\log(1/z) + \gamma}{z} - \frac{1}{4} + O(z), \qquad \mathcal{F}_0''(z) = O(W/t^2).$$

The disk stays in the right half-plane. Taylor's theorem then shows that on its boundary $|\mathcal{F}_0(z) - Y| \ge c_1 W t$, and that $\mathcal{F}_0(z) - Y$ has one simple zero inside. The exponential integral representation gives uniformly

$$|J(z)| \le c_2 t e^{-A/t},$$

since $1/z = 1/t + O(1)$ throughout the disk; the arithmetic tail is summable uniformly there. Put $\rho(t) = \exp(3A/(4t))$. Then $\rho(t)|J(z)| = O(te^{-A/(4t)}) = o(Wt)$. Rouché's theorem shows that for every $|\eta| \le \rho(t)$ the perturbed equation has exactly one zero in the disk, counted with multiplicity. It is therefore simple, and the implicit function theorem makes it holomorphic in η. In particular its Taylor series converges at $\eta = \pm 1$ for small enough t.

To compute that Taylor series, use the local coordinate $v = \mathcal{F}_0(z)$ and write $z = g_0(v)$, extending the inverse locally in Y. Equation (8.1) becomes

$$v = Y - \eta h(v), \qquad h(v) = J(g_0(v))/2.$$

Lagrange–Bürmann applied to the analytic function $g_0(v)$ gives (8.2). Taking its odd and even parts in η gives the other two identities. This argument uses ordinary local analytic inversion in an auxiliary parameter, not an assumed convergence of the factorial q-sector series. □

8.2 The first two inverse Stokes effects

Theorem 8.2 (Nested jump and exact leading averaging defect)**.** *For $t = g_0(Y) \to 0^+$, $W = \log(1/t) + \gamma$, and $A = 4\pi^2$,*

$$g_+(Y) - g_-(Y) = \frac{4\pi \mathrm{i}}{A}\frac{t^2}{W}e^{-A/t}\{1 + O(t)\}, \tag{8.5}$$

$$\frac{g_+(Y) + g_-(Y)}{2} - g_0(Y) = -\frac{t^2}{W^2}e^{-2A/t}\{1 + O(t)\}. \tag{8.6}$$

In the physical variables of (7.1),

$$g_+(Y) - g_-(Y) = \frac{\mathrm{i}}{\pi}\frac{q^2}{X}\exp\left(-\frac{A}{q} - \frac{A}{4X}\right)\{1 + O(q)\}, \tag{8.7}$$

$$\frac{g_+(Y) + g_-(Y)}{2} - g_0(Y) = -\frac{q^2}{X^2}\exp\left(-\frac{2A}{q} - \frac{A}{2X}\right)\{1 + O(q)\}. \tag{8.8}$$

Thus inversion exposes a doubly exponential Stokes level, with leading action $A/q = Ae^{X-\gamma}$.

Proof. The exact forward expansion and (5.7) imply

$$g_0'(Y) = -\frac{t}{W}\{1 + O(t/W)\}, \qquad J(t) = \frac{4\pi \mathrm{i}}{A}te^{-A/t}\{1 + O(t)\}. \tag{8.9}$$

The first term of (8.3) is $-g_0'(Y)J(t)$, giving (8.5). For rigor in discarding the remaining terms, the proof of Theorem 8.1 bounds $g_\eta(Y) - t$ by ct^2 on $|\eta| = \rho(t)$. Its Taylor coefficients are therefore bounded by $ct^2\rho(t)^{-k}$. The remaining odd terms begin at $k = 3$, and are $O(t^2e^{-9A/(4t)})$, which is negligible relative to the first term.

The first term of (8.4) is

$$\frac{1}{8}\partial_Y\{g_0'(Y)J(t)^2\}. \tag{8.10}$$

The remaining even terms begin at $k = 4$ and are $O(t^2e^{-3A/t})$, again negligible. Since $\partial_Y = g_0'(Y)\partial_t$, the leading product inside (8.10) is

$$\frac{16\pi^2}{A^2}\frac{t^3}{W}e^{-2A/t}.$$

Differentiating its exponential and multiplying by $-t/W$ gives

$$-\frac{32\pi^2}{A}\frac{t^2}{W^2}e^{-2A/t}.$$

Division by 8 leaves $-4\pi^2/A = -1$. Derivatives of the algebraic prefactor, and the differentiated remainder terms, produce a relative $O(t)$ error. These derivative estimates follow either by integration by parts in E_1 with differentiated remainders or by its exact derivative; they are not obtained by differentiating an arbitrary uncontrolled O term. This proves (8.6).

Finally Theorem 7.4 gives $t = q\{1 - q/(4X) + O(q^2/X)\}$, $W = X + O(q/X)$, and $A/t = A/q + A/(4X) + O(q/X)$. Substitute these expressions in the two preceding formulas. The exponent error is $O(q/X)$, so it can be exponentiated into a relative $1 + O(q)$ factor, proving (8.7)–(8.8). □

Remark 8.3 (A normalization safeguard). Replacing t merely by q *inside* $e^{-A/t}$ loses the specific algebraic exponential multiplier $e^{-A/(4X)}$. Although that multiplier tends to 1, it matters when resolving successive slow corrections within the same nested action. At the same time, arithmetic averaging of g_+ and g_- is not the same operation as inverting the arithmetic average of $\mathcal{F}_+$ and $\mathcal{F}_-$. Equation (8.6) quantifies that difference. This does not contradict multiplicativity properties of a properly defined nonlinear median summation operator: no such operator is being identified with a plain arithmetic average of the inverse branches.

9 Verification and reproducibility

The proofs above are independent of the verification program. The program is intended to catch normalization, sign, indexing, branch, and transcription errors before integration.

9.1 Exact checks

The SymPy portion constructs (7.5) through $n = 8$ using rational zeta values. It then substitutes the resulting polynomials into (7.3) by an independent recurrence for the coefficients of e^{-mV}:

$$E_0 = 1, \qquad E_n = -\frac{m}{n}\sum_{j=1}^{n} jC_jE_{n-j}.$$

Every residual coefficient through q^8 is checked to be exactly zero. The program also checks the polynomial degrees, valuations, all coefficient signs, the Catalan leading coefficients, and the odd-sector u^2 coefficients. A separate symbolic multiplication checks a_0, a_1, a_2, a_3 in the centered normalizer. These are finite exact checks, not formal proofs of the infinite family.

9.2 Numerical diagnostics

The mpmath portion compares the two sides of the reflected coefficient identity at real and complex orders; checks the second-jet formula at three odd indices; verifies pole coalescence; solves the scaled core equation and its two-variable critical equations; and compares the forward asymptotic expansion with the absolutely convergent Li_1 sum. An optional SciPy calculation independently compares (5.6) with the original Lambert sum at $t = 1$ and $t = 3$. It truncates the vertical integral at height 55 and records the adaptive quadrature error estimate. That estimate is not an interval certificate.

For the most delicate inverse checks, the program solves $\mathcal{F}_0(z) + J(z)/2 = \mathcal{F}_0(t)$ at $t = 1, 1/2, 1/4$ with increased precision. Newton iteration uses the independently differentiated convergent series

$$\mathcal{F}_0'(t) = -\sum_{a\geq 1}\frac{e^{-at}}{1-e^{-at}}, \qquad J'(t) = \frac{4\pi\mathrm{i}}{t}\sum_{n\geq 1} d(n)e^{-An/t}. \tag{9.1}$$

The conjugate root is the lower lateral inverse. The recorded ratios test both the first-action imaginary splitting and the much smaller second-action real bias. Near this nested scale, a

default generic secant solver is not a reliable residual certificate; explicit Newton steps and a final residual check are used instead.

t	Working digits	Jump / leading term	Bias / leading term
1	80	0.686714522854	0.494775281614
1/2	125	0.900289105501	0.829028586173
1/4	205	0.963296642480	0.938079390839

These are ratios to the asymptotic leading terms in (8.5) and (8.6), not relative errors in the exact identities. Their approach to 1 is a diagnostic for the stated small-t laws. A recorded zero root residual at finite working precision means only that the residual rounded to zero; it is not an exact certificate.

9.3 How to reproduce

From the package directory, run

```
python verification/verify.py --output verification/results.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The file `verification/results.json` records the actual run, including software versions and the scope of each check. The flag `-skip-lateral` omits only the higher-precision lateral-root diagnostics. SciPy is optional; without it, the independent Mellin quadrature is explicitly marked as not run. No network access is needed by the verification program or the TeX build.

10 Source audit, corrections, and integration boundaries

10.1 What was inspected

The repository was anchored at commit `6207c3d40047fd8d1400a4d509fee71c9a3a8132`, whose recorded message was “New research reports.” The inspection included the Polylogarithms overview, the Transseries overview and its series-and-transseries overview, the incoming directory inventory and README, the manuscript chapter inventory, and the normalized positive–negative derivative-bridge chapter. Three library research drafts were used for an abstract-level comparison and selected statement checks. The ZIP byte contents were not downloaded; no byte-identity claim between those library drafts and individual ZIP members is made. This is a targeted research audit, not a validation of the entire repository.

10.2 No unsupported source-specific correction

No new erroneous theorem in the inspected repository material has been established by this work. In particular, the convergence claims in the selected prior drafts concern holomorphic analytic tails and do not conflict with Theorem 7.2. We therefore provide no patch that weakens or rewrites those existing theorems.

Two *proposed integration safeguards* are mathematically substantive. First, replacing a single polylogarithm by the Lambert superposition (2.2) destroys the analyticity of the ordinary tail at the origin: (4.7) proves radius zero. Any extension that asserted an ordinary convergent inverse-sector series for this new family at fixed $u > 0$ would be false, and Theorem 7.2 supplies its correct Gevrey replacement. Second, interpreting the arithmetic mean of the lateral inverse branches as the real inverse would be false, with the explicit nonzero counterterm (8.6). Neither warning is attributed to an inspected source that did not make the offending claim.

10.3 Recommended integration shape

The contribution is designed as a standalone research package, with cross-links from Polylogarithms and Transseries. A suitable topic label is `resonant-lambert-inverse-transseries`. The Borel and Mellin statements should cite the established Lambert literature rather than replace it. The new package should retain separate status labels for conventional proofs, exact finite checks, ordinary numerical diagnostics, and future proof-assistant work. No theorem here has been compiled in Lean, and no numerical output is presented as a rigorous interval enclosure.

The `AUDIT.md`, `PROVENANCE.md`, and `CLAIMS.md` files summarize these distinctions outside the TeX article. The package does not alter the canonical manuscript or publish any repository commit.

11 Further research questions

The following problems are concrete continuations, not premises used in the preceding proofs.

11.1 Sharp large-order constants of the inverse sectors

Theorem 7.2 fixes the exponential Borel scale but not the full large-order coefficient law. A natural conjecture, for fixed $u > 0$, is

$$C_{2m}(u) \sim -\frac{4u\Gamma(2m)}{2mA^{2m}}\cosh(Au/4), \tag{11.1}$$

$$C_{2m+1}(u) \sim \frac{4u\Gamma(2m+1)}{(2m+1)A^{2m+1}}\sinh(Au/4). \tag{11.2}$$

These are *unproved in this article*. For each fixed number of factors in (7.5), the contribution of one large even index and the remaining linear indices leads to the displayed hyperbolic series. A proof must control the sum over the number of factors uniformly with m; the current $e^{K\sqrt{n}}$ upper bound is not sufficient for that interchange. The conjecture would distinguish the amplitudes at $+A$ and $-A$ and sharpen the reversion calculus.

11.2 Full inverse Borel continuation

Does $\widehat{V}_u$ continue along every path avoiding an explicit additive arithmetic action set? Can one describe its positive-axis singularity directly from (7.5), rather than from analytic inversion of the forward lateral functions? The answer would turn the present exact transport identities into a more complete resurgent classification of the formal inverse itself.

11.3 Uniform Stokes transport away from exact resonance

The hypergeometric kernel (4.5) is jointly analytic in the order parameter. Deriving its full discontinuity in closed hypergeometric coordinates, and then resolving the scaled regime $\varepsilon = \lambda/X$, should connect the arcsinh crossover to moving arithmetic Stokes amplitudes. All branch choices and the distinction between core and full function must remain explicit.

11.4 From core folds to full complex Lambert folds

Theorem 6.3 concerns only the singular core. One may ask whether suitable sectors permit a full-function deformation of those critical loci, and whether the resulting inverse is uniformly described by a quadratic local normal form with an exponentially small arithmetic displacement. Direct substitution into a complex scaled path is not enough: the full Lambert sum must remain in a domain of analytic continuation with controlled remainders.

11.5 Character pairs and higher collision multiplicity

The two-character family in [1] permits the same inverse questions with twisted divisor weights. Primitive character parities should determine the vanishing order of perturbative sectors. Another extension is the product-lattice sum over r positive integers, whose Mellin kernel contains r zeta factors. At equal indices it has an $(r+1)$-fold pole and an r-th degree logarithmic core. Its inverse would naturally introduce $e^{-cY^{1/r}}$ sectors and nested Stokes levels. Explicit centered order-jet formulas are a tractable first step.

11.6 Certified computation and formal proof organization

An interval implementation of the logarithmic Borel integral would provide a practical check of exponentially small corrections without subtraction of enormous nearly equal terms. On the formal side, the coefficient formula, sign theorem, Catalan degree, and finite composition estimates are natural initial targets. Mellin contour shifts, parameter-uniform asymptotics, and analytic Stokes transport form a separate, substantially larger analytic formalization task.

Conclusion

The double Lambert specialization connects classical polylogarithms, Gamma–Stieltjes order calculus, arithmetic Borel singularities, and nonlinear transseries inversion in a single exactly normalized example. The triple pole creates a quadratic logarithmic core; its scaled inverse has an arcsinh crossover. Trivial-zero parity produces a full missing forward sector, restored both by the second order jet and, through a different mechanism, by nonlinear reversion. The reversion coefficients have an exact rational formula, Catalan top degree, alternating signs, and a sharp Borel scale despite ordinary divergence. Finally, the exact arithmetic jump survives inversion as a nested exponential hierarchy, and the first real averaging defect has the particularly simple coefficient -1. These results provide a focused bridge between the two repository areas while keeping classical input, proposed refinements, proofs, tests, and unresolved extensions separate.

References

[1] D. Broadhurst and D. Dorigoni, *Resurgent Lambert series with characters*, arXiv:2507.21352, version 1 (28 July 2025), especially Sections 2.3 and 3. https://arxiv.org/html/2507.21352v1.

[2] D. Broadhurst and D. Dorigoni, *Resurgent Lambert series from Feynman and beyond*, arXiv:2607.14020v1 (15 July 2026), Section 4, equations (30)–(31). https://arxiv.org/html/2607.14020v1.

[3] D. Dorigoni and A. Kleinschmidt, *Resurgent expansion of Lambert series and iterated Eisenstein integrals*, Communications in Number Theory and Physics **15** (2021), 1–57; arXiv:2001.11035. https://arxiv.org/abs/2001.11035.

[4] NIST Digital Library of Mathematical Functions, Section 25.12, *Polylogarithms*, especially equation 25.12.10. https://dlmf.nist.gov/25.12.

[5] NIST Digital Library of Mathematical Functions, Section 25.4, *Reflection Formulas* for the Riemann zeta function. https://dlmf.nist.gov/25.4.

[6] NIST Digital Library of Mathematical Functions, Section 5.7, *Series Expansions* for the Gamma function. https://dlmf.nist.gov/5.7.

[7] V. Reshetnikov, *ProveIt: Analysis/Polylogarithms*, repository overview, inspected 12 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/6207c3d40047fd8d1400a4d509fee71c9a3a8132/Analysis/Polylogarithms.

[8] V. Reshetnikov, *ProveIt*, manuscript chapter `08-bridges.tex`, "One normalized functional equation for all derivative bridges," inspected at the same commit. https://github.com/VladimirReshetnikov/ProveIt/blob/6207c3d40047fd8d1400a4d509fee71c9a3a8132/Analysis/Polylogarithms/docs/manuscript/chapters/08-bridges.tex.

[9] V. Reshetnikov, *ProveIt: Analysis/Transseries*, overview and `docs/series-and-transseries/README.md`, inspected 12 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/6207c3d40047fd8d1400a4d509fee71c9a3a8132/Analysis/Transseries.

[10] V. Reshetnikov, *ProveIt: docs/incoming*, inventory and README, inspected at the same commit. https://github.com/VladimirReshetnikov/ProveIt/tree/6207c3d40047fd8d1400a4d509fee71c9a3a8132/docs/incoming.

[11] *Resonant Polylogarithmic Jets and Convergent Inverse Transseries: Gamma normal forms, optimal dilation filters, arithmetic sector selection, and exact order tuning*, research draft prepared for V. Reshetnikov, 11 October 2026. Library source `resonant_polylog_transseries.tex`; abstract and selected normal-form statements inspected.

[12] *Resonant Polylogarithm Inversion: Uniform Order Crossover, Gamma–Zeta Fold Loci, and Convergent Exponential Sectors*, research draft prepared for V. Reshetnikov, 11 October 2026. Library source `article(20261012-045040).tex`; abstract and selected core statements inspected.

[13] *Resonant Polylogarithm Transseries: Gamma Borel Kernels, Cyclotomic Order Jets, and Sharp Dilogarithm Reversion*, research draft prepared for V. Reshetnikov, 11 October 2026. Library source `article(20261012-044824).tex`; abstract and selected convergence statements inspected.